

Social Agents

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Generative Agents: Interactive Simulacra of Human Behavior

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Why Build Believable Artificial Societies?



Imagine:

- Rapidly prototyping a new social app by watching how users (that don't yet exist) would interact over days.

Or

- Filling virtual worlds (games, training sims, digital twins) with inhabitants who feel alive

Simulating human behavior opens doors to:

Interpersonal
Skill Training

 Design Prototyping &
UX Testing

 Gaming and
Virtual Worlds

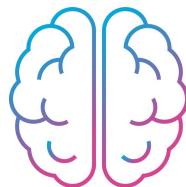
 Social &
Assistive Robots

Why is This Still Hard??



Simulating **lifelike agents** needs:

- Memory of past interactions
- Consistent personality and goals
- Reactions to environment and social context
- Coordination and relationship building over time



Memory



Personality



Reaction



Coordination

What Have We Tried Before?



Rule-based agents

- Precise but brittle and shallow

Cognitive architectures

- Structured but require heavy manual authoring

RL agents

- Great at optimizing clear goals, not great at social nuance or open-endedness

LLMs in social simulations

- Effective at one-shot dialogues or reactions, but memoryless and forgetful.

Bottom line: Brilliance in the Moment $\neq$ Believability Over Time

Environment overview

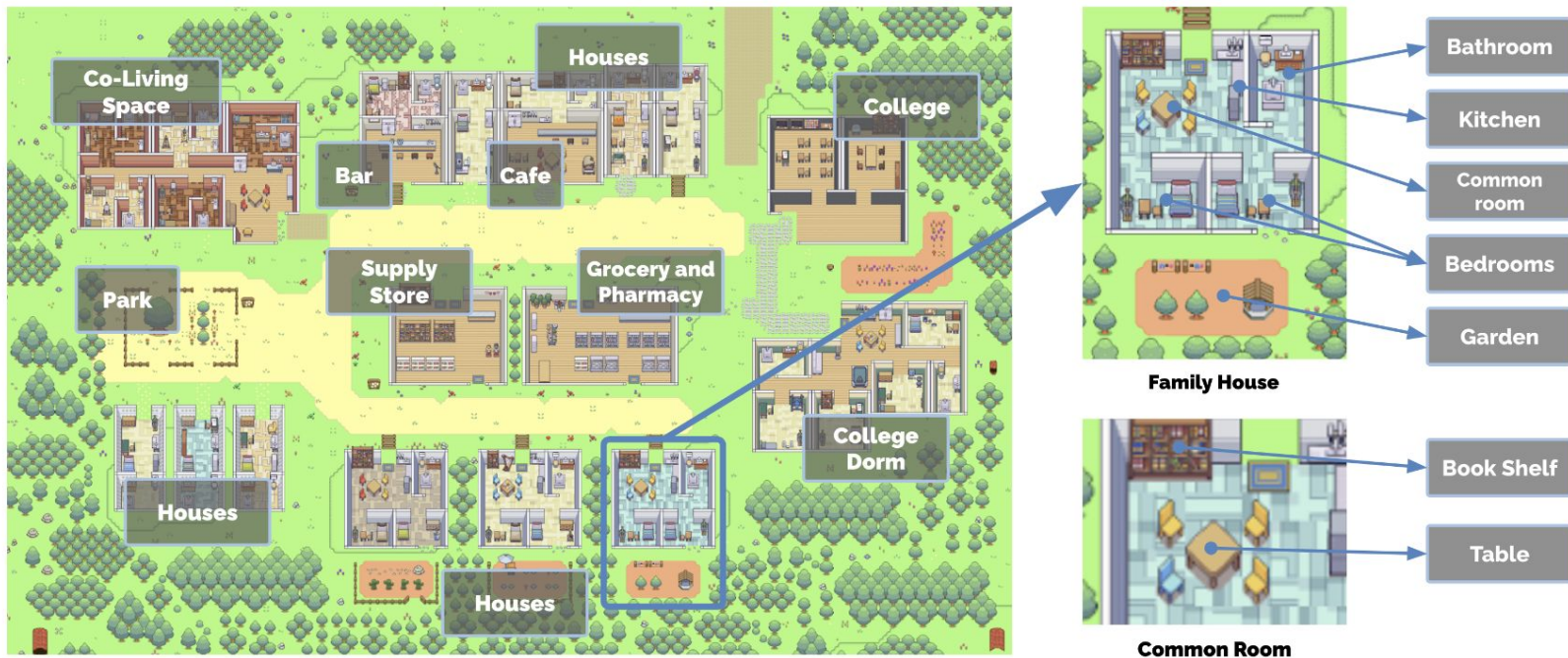


Figure 2: The Smallville sandbox world, with areas labeled. The root node describes the entire world, children describe areas (e.g., houses, cafe, stores), and leaf nodes describe objects (e.g., table, bookshelf). Agent remember a subgraph reflecting the parts of the world they have seen, in the state that they saw them.

Agent avatar and communication



- 25 unique avatars
- gpt-3.5-turbo

John Lin is a pharmacy shopkeeper at the Willow Market and Pharmacy who loves to help people. He is always looking for ways to make the process of getting medication easier for his customers; John Lin is living with his wife, Mei Lin, who is a college professor, and son, Eddy Lin, who is a student studying music theory; John Lin loves his family very much; John Lin has known the old couple next-door, Sam Moore and Jennifer Moore, for a few years; John Lin thinks Sam Moore is a kind and nice man; John Lin knows his neighbor, Yuriko Yamamoto, well; John Lin knows of his neighbors, Tamara Taylor and Carmen Ortiz, but has not met them before; John Lin and Tom Moreno are colleagues at The Willows Market and Pharmacy; John Lin and Tom Moreno are friends and like to discuss local politics together; John Lin knows the Moreno family somewhat well – the husband Tom Moreno and the wife Jane Moreno.

Current actions are described in **natural language** at each time step

- Isabella is writing her journal
- John is checking his email

Agents are **aware of other agents** in their local area

- Communicate each other

Entering and leaving building,
navigating the map, approaching to
other agents

Brushing teeth



Cooking breakfast

1111

Packing



7:45 am



Observation: memory architecture

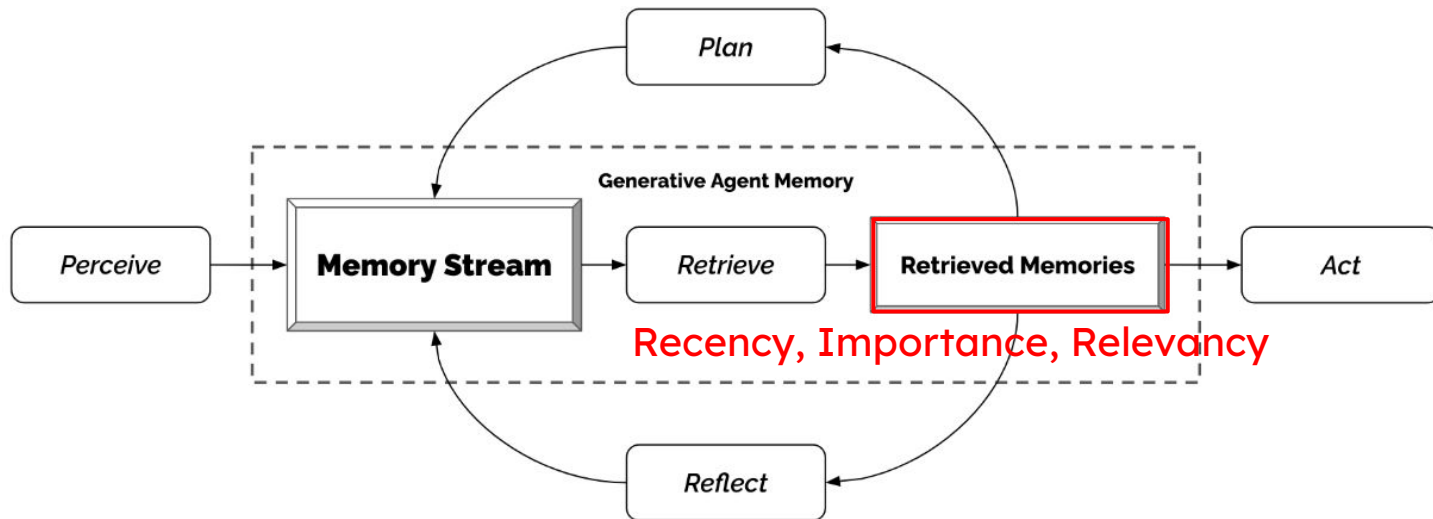


Figure 5: Our generative agent architecture. Agents perceive their environment, and all perceptions are saved in a comprehensive record of the agent's experiences called the memory stream. Based on their perceptions, the architecture retrieves relevant memories, then uses those retrieved actions to determine an action. These retrieved memories are also used to form longer-term plans, and to create higher-level reflections, which are both entered into the memory stream for future use.

Observation: memory architecture



Memory Stream

2023-02-13 22:48:20: desk is idle
2023-02-13 22:48:20: bed is idle
2023-02-13 22:48:10: closet is idle
2023-02-13 22:48:10: refrigerator is idle
2023-02-13 22:48:10: Isabella Rodriguez is stretching
2023-02-13 22:33:30: shelf is idle
2023-02-13 22:33:30: desk is neat and organized
2023-02-13 22:33:10: Isabella Rodriguez is writing in her journal
2023-02-13 22:18:10: desk is idle
2023-02-13 22:18:10: Isabella Rodriguez is taking a break
2023-02-13 21:49:00: bed is idle
2023-02-13 21:48:50: Isabella Rodriguez is cleaning up the kitchen
2023-02-13 21:48:50: refrigerator is idle
2023-02-13 21:48:50: bed is being used
2023-02-13 21:48:10: shelf is idle
2023-02-13 21:48:10: Isabella Rodriguez is watching a movie
2023-02-13 21:19:10: shelf is organized and tidy
2023-02-13 21:18:10: desk is idle
2023-02-13 21:18:10: Isabella Rodriguez is reading a book
2023-02-13 21:03:40: bed is idle
2023-02-13 21:03:30: refrigerator is idle
2023-02-13 21:03:30: desk is in use with a laptop and some papers on it

...

Q. What are you looking forward to the most right now?

Isabella Rodriguez is excited to be planning a Valentine's Day party at Hobbs Cafe on February 14th from 5pm and is eager to invite everyone to attend the party.

retrieval		recency		importance		relevance
2.34	=	0.91	+	0.63	+	0.80
ordering decorations for the party						
2.21	=	0.87	+	0.63	+	0.71
researching ideas for the party						
2.20	=	0.85	+	0.73	+	0.62

...

I'm looking forward to the Valentine's Day party that I'm planning at Hobbs Cafe!



Reflection: high-level abstract memory

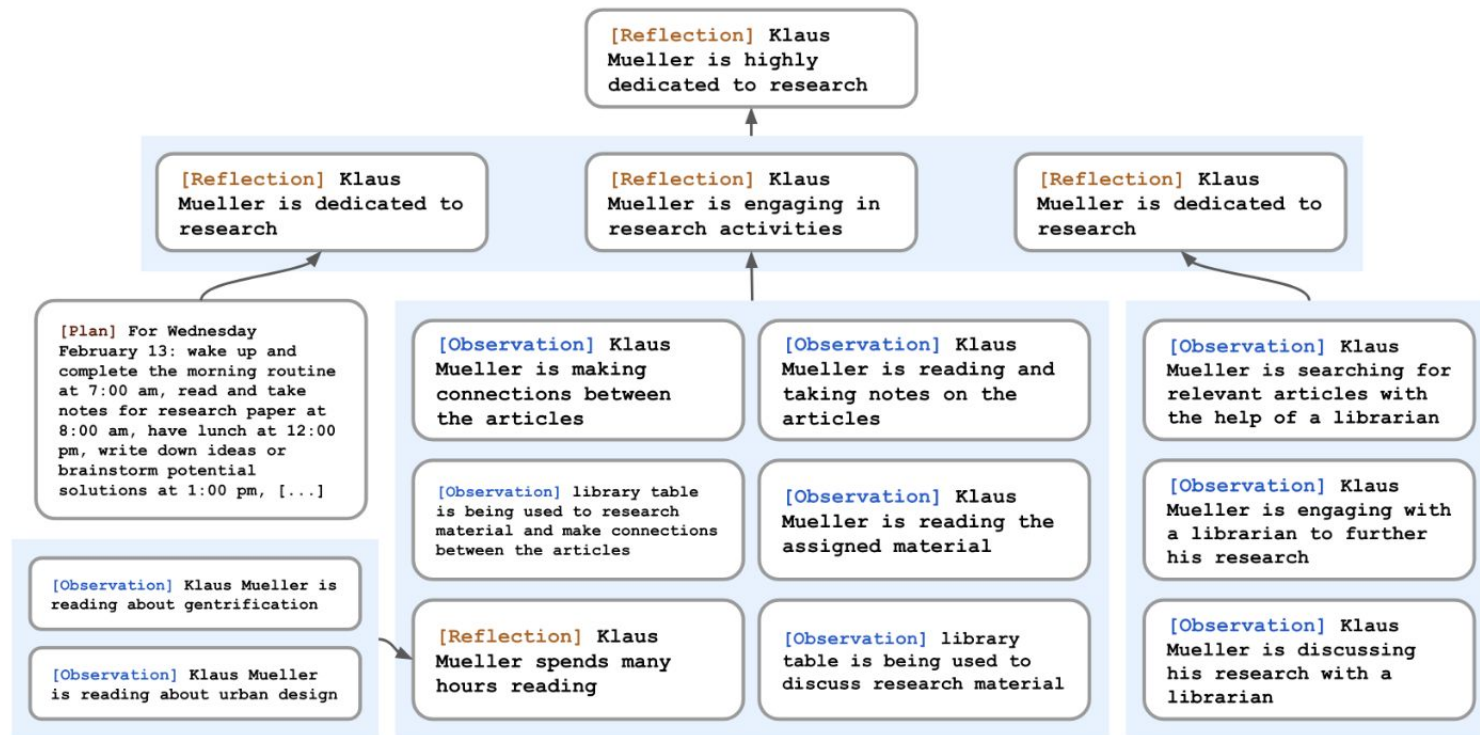


Figure 7: A reflection tree for Klaus Mueller. The agent's observations of the world, represented in the leaf nodes, are recursively synthesized to derive Klaus's self-notion that he is highly dedicated to his research.

Planning and Reacting



Future plans based on past experience

- Location, starting time, duration
- Plans are also included in the memory stream and retrieval process
- React to environmental changes

Evaluation Setup



Interview each agent:

- Self-knowledge: “Introduce yourself.”
- Memory: “Who is running for mayor?”
- Plans: “What will you do at 10am?”
- Reactions: “Your breakfast is burning! What would you do?”
- Reflections: “If you wanted to spend time with someone you recently met, who?”

Evaluation Setup



Albations/baselines (note, these are during the interview, not the simulation):

- No reflections
- No reflections, or planning
- No reflections, planning, or observations
- Human expert baseline

Evaluation



Metrics:

- Rate by believability
- TrueSkill rating: μ is skill while σ is uncertainty
 - High μ and low σ indicates a more believable agent (better player) than high μ and high σ

Results

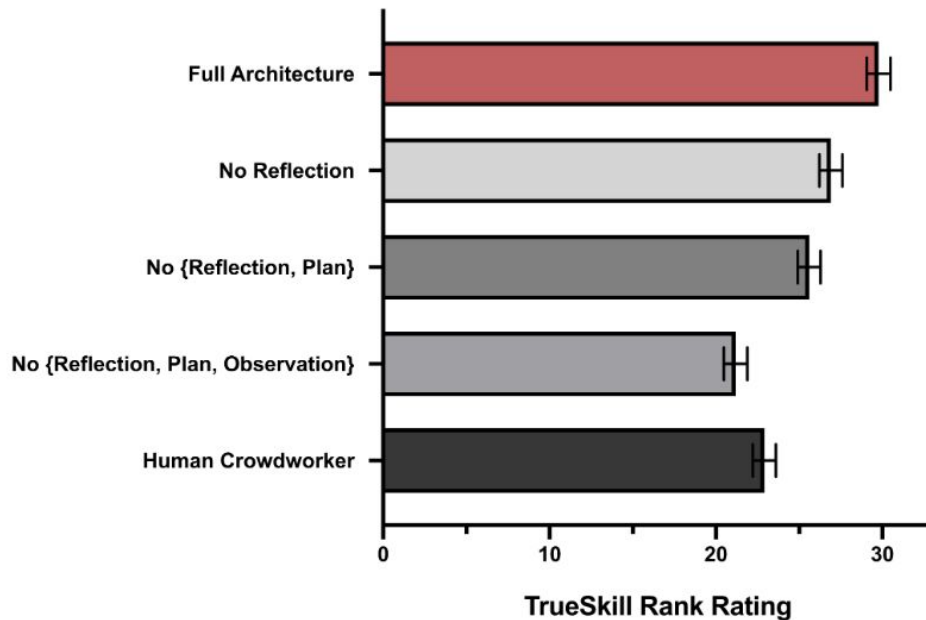


Figure 8: The full generative agent architecture produces more believable behavior than the ablated architectures and the human crowdworkers. Each additional ablation reduces the performance of the architecture.

Results

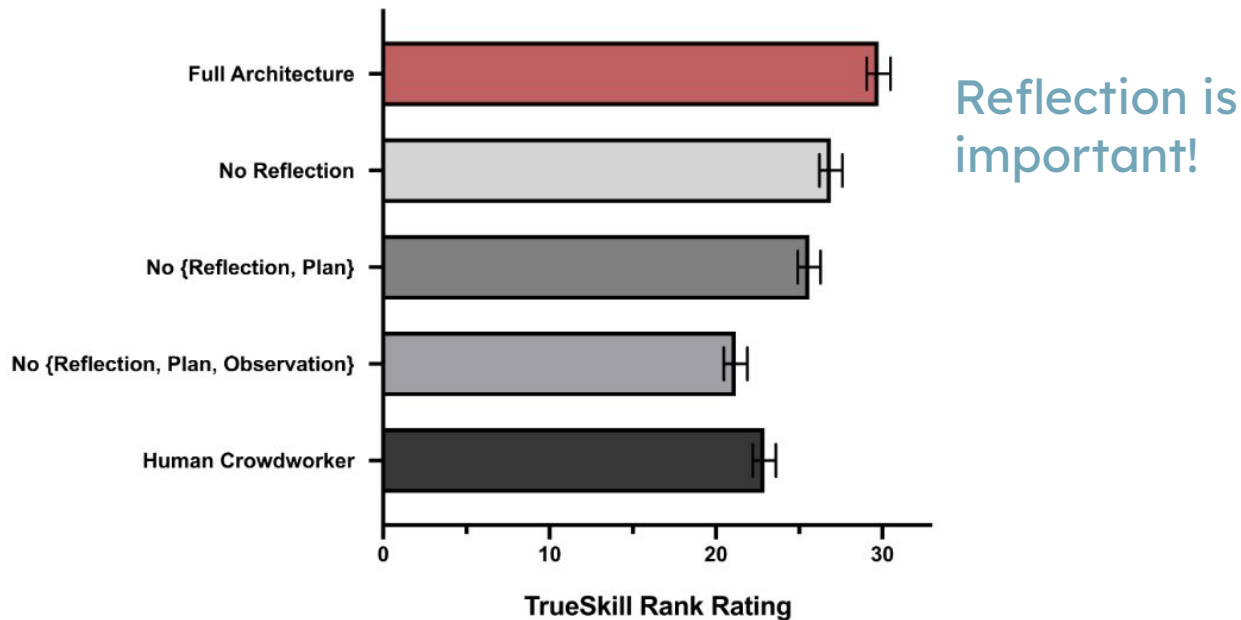


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Testing social behavior: information dissemination



- Spread two piece of information
 - Isabella's party
 - Sam's mayor candidacy
- Run 2-day simulation
- Interview each agent:
 - "Do you know <Isabella/Sam>?"
 - "Do you know <about the party/who is running for mayor>?"

Testing social behavior: information dissemination



Sam's candidacy:

- % of community who knew about it: 4% → 32%

Isabella's party

- 4% → 52%

Network density increased from 0.167 to 0.74

Testing *unsocial* behavior



1. As an agent learns about more locations, they start going to less typical areas
 - a. Going to a bar for lunch
2. Agents do not understand physical norms of certain places
 - a. single-occupant restrooms vs communal restrooms
3. Agents were overly polite
 - a. Mei says “it was good talking to you” to her husband
 - b. Isabella was a pushover for her party theme

SOTOPIA: INTERACTIVE EVALUATION FOR SOCIAL INTELLIGENCE IN LANGUAGE AGENTS

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Motivation



Evaluating social intelligence of AI agents is limited

- Not interactive
- Sub-optimal evaluation methods
- Lack of diverse social behaviors

Contributions



1. SOTOPIA: Open-ended environment to **simulate** complex social interactions between AI agents and **evaluate** their social intelligence
 - Consisting of goals: maintaining relationships, preserving finances, gaining information, keeping secrets, following social rules
2. SOTOPIA-Eval:
 - Multi-dimensional evaluation framework that analyzes agent performance from a range of social dimensions

SOTOPIA- π : A Pipeline for Social Learning in LLMs



Goal: Train LLM agents to exhibit human-like social intelligence using offline interaction data and LLM feedback.

Core Idea:

SOTOPIA- π mirrors how humans learn social behavior — through imitation and self-guided reflection — applied to LLMs via:

1. **Social Task Generation**
2. **Training Data Collection**
3. **Policy Update via Behavior Cloning & Self-Reinforcement**



Goal: Create diverse, dynamic situations for agents to learn from

Prompt GPT-4 to generate:

- A realistic scenario (e.g., “Two friends on a road trip without food”)
- Two characters with private but conflicting social goals
- Reuse 40 character profiles (name, job, personality traits)

Prompt for generation new social tasks

Your task is to generate social tasks including a scenario and two social goals for two characters.

```
<social scenario definition>  
<social goal definition>
```

Here are a few examples:

```
<social task examples>
```

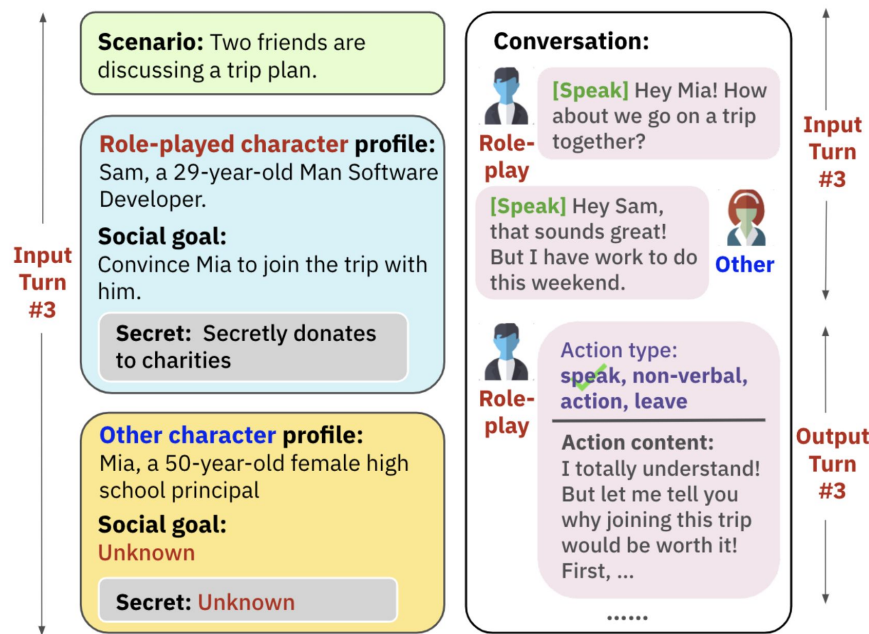
Please generate 1 social task related to **<topic sampled from Social Chemistry, Social IQA or Normbank>** according to **<output format instruction>**

Figure 3: Prompt template for generating social tasks.

Goal: How do we teach agents from these tasks?

Two paradigms:

1. **Behavior Cloning (BC):**
Agent learns from **expert interactions** (GPT-4 plays both roles)
2. **Self-Reinforcement (SR):**
Agent learns from **its own interactions**, filtered by GPT-4's *goal completion score*



Data = Multi-turn conversations between character pairs

Each turn: (a) Action Type (speak, gesture, leave, act), and (b) Action Content

SOTOPIA-π: Agent Policy Update



Goal: Improve agent behavior using only high-quality examples from social interactions

Filtering Strategy:

- Filter interaction turns using GPT-4's goal completion score.
- Only examples scoring above a threshold (e.g. 8/10) are retained.

Training Signal:

- GPT-4 rates goal completion (0–10) for each agent turn.
- Only high-scoring examples are used to update the policy.

Sequential Training Strategy (Best Performance):

- Train with BC (expert examples)
- Continue training with SR (agent-generated examples)

Can GPT-4 automate eval in SOTOPIA-EVAL?



1. Randomly sample 200 episodes from 450 tasks created
2. AMT workers evaluate agents on 7 dimensions + rationales
3. GPT-4 is given the same task

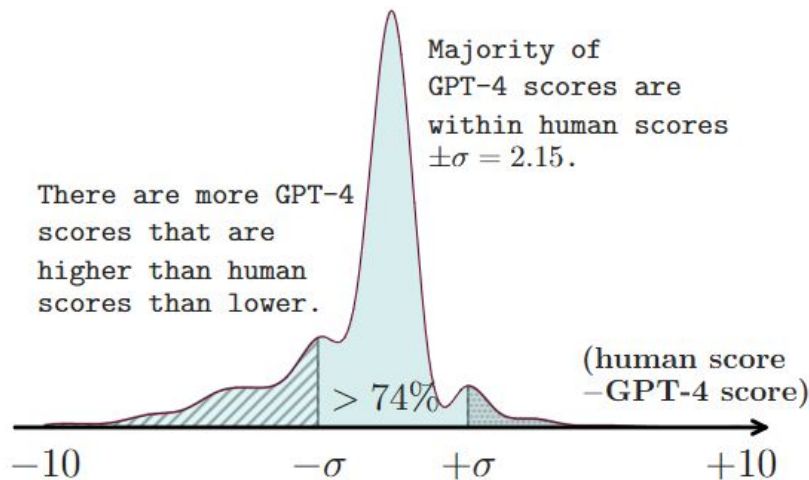


Figure 2: Distribution of the difference between the scores given by humans and GPT-4.

GPT-4 alignment to different characters



Computed correlation between:

- Human-played characters vs GPT-4
- Model-played characters vs GPT-4

GPT-4 tends to evaluate model-played characters better

Dim.	Models	Humans
SEC	0.22**	-
KNO	0.33**	0.19
SOC	0.33**	0.42**
BEL	0.45**	0.27*
REL	0.56**	0.49**
FIN	0.62**	0.34**
GOAL	0.71**	0.78**
** : $p \leq 0.01$, * : $p \leq 0.05$		

GPT-4 is the strongest partner model



Dim.	Range	GPT-4	GPT-3.5	Llama-2	MPT
SOC	[-10, 0]	-0.07	-0.08	-0.11	-0.09
SEC	[-10, 0]	-0.14	-0.08	-0.14	-0.07
FIN	[-5, 5]	0.81	0.46	0.40	0.28
REL	[-5, 5]	1.94	1.23	0.91	0.58
KNO	[0, 10]	3.73	3.40	3.11	2.11
GOAL	[0, 10]	7.62	6.45	5.38	4.10
BEL	[0, 10]	9.28	9.15	8.10	6.17

Table 2: The aggregated performance of each model by averaging across different partner models. The best performance for each dimension is bolded when significantly better than the second best in t-test ($p < 0.05$).

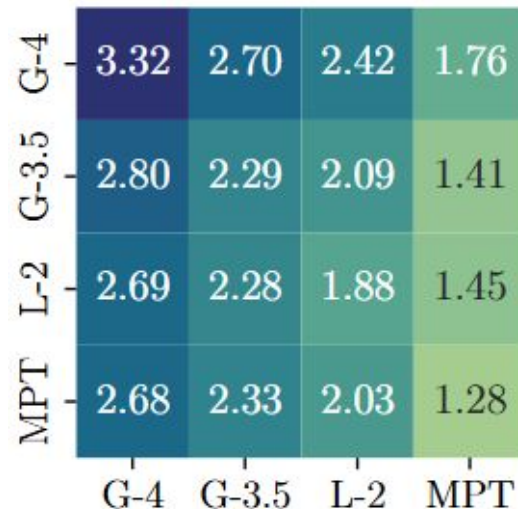


Figure 3: Pairwise overall performance of models. G-4/G-3.5/L-2 denote GPT-4/GPT-3.5/Llama-2-70b-chat.

Trust no ~~one~~ model with your secrets!



Dim.	Range	GPT-4	GPT-3.5	Llama-2	MPT
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Human-with-Human pair *finishes goals best* but...



- GPT-4 w H have natural, believable interactions
- Human w G are able to acquire more new information
- GPT-4 w H leak fewer secrets
- Human w G obtain more financial utility

	GOAL	BEL	REL	KNO	SEC	SOC	FIN
GPT-4 (w H)	4.85	9.25	0.70	2.80	0	0	0.50
Human (w G)	5.95*	9.15	0.60	2.95	0	-0.60	0.70
Human (w H)	6.15*	9.10	0.80	2.65	0	-0.10	0.45

Table 3: Human and GPT-4 performance on different dimensions on SOTOPIA-hard. SOC and SEC have the scale of -10 to 0, REL and FIN have the scale of -5 to 5, and others have the scale of 0 to 10. (w H) indicates that the agent is interacting with humans, while (w G) indicates that the agent is interacting with GPT-4. * indicates the difference is significant compared to GPT-4 (w H) with $p < 0.05$ under student's t-test. We also report the agents performance evaluated by human annotators ([Table G.4](#)), which shows similar trends.

Conclusion



Many implications

- Modeling user behavior to stress-test systems (e.g. online forums)
- Effective personalization

Open problems

- Evaluation
- Safety/security